higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N4

(16030164)

18 November 2019 (X-Paper)
09:00–12:00

Scientific calculators may be used.

This question paper consists of 5 pages and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL intermediate steps and simplify where possible.
5. ALL final answers must be rounded off to THREE decimal places (unless indicated otherwise)
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Use only a BLUE or BLACK pen.
8. Write neatly and legibly.

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QUESTION 1

1.1 1.1.1 Sketch the graph of $y = -\sqrt{9-x^2}$ (3)

1.1.2 Is the graph above a function or a relation? (1)

1.2 1.2.1 Sketch the graph of $\log_2 y = x$ and its inverse on the same set of axes. (4)

1.2.2 With respect to which line are the graphs in QUESTION 1.2.1 asymmetrical? (1)

1.2.3 Is the graph of the inverse of $\log_2 y = x$ continuous? (1)

1.3 Solve for x if:

$$x^2 + 2x = -2$$ (4)

1.4 Simplify the following and leave answer in $a + jb$ form:

$$z = \frac{\left[8\left(\cos\frac{\pi}{6} + j\sin\frac{\pi}{6}\right)\right]^3}{(5cis\,45°)^3}$$ (4)

1.5 Simplify: j^3 (2)

[20]

QUESTION 2

2.1 Determine the value of $\sin 75°$ without the use of a calculator. (3)

2.2 Show that $\tan(270° - x) = \cot x$ (3)

2.3 Solve for:

$$5\tan^2 x + 6 = 11\tan x; 0° \le x \le 360°$$ (6)

2.4 Simplify:

$$\frac{\sin(300°).\sec(150°) + \tan(225^0)}{\cos(300°) - \sin(210°)}$$ (3)

2.5 Prove that:

$$\frac{\cos x}{1-\cos x} + \frac{\cos x}{1+\cos x} = 2\cot x.\cos ecx$$ (5)

[20]

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QUESTION 3

3.1 3.1.1 If $y = x^3 - 3x$ prove that the minimum turning points are (1;-2) and the maximum turning points are (-1;2). Show ALL calculations. (7)

3.1.2 Calculate the points of inflection of $y = x^3 - 3x$. (2)

3.2 Use the product rule to differentiate $y = 3x.\sin 3x$. (4)

3.3 3.3.1 Sketch and clearly indicate the area bounded by the graph of $y = x$, the X-axis; $x = -1$ and $x = 1$. Also, show the representative strip used to calculate the area. (3)

3.3.2 Determine, using integration, the value of the area shown in QUESTION 3.3.1. (4)

[20]

QUESTION 4

4.1 Differentiate the following in terms of x:

$$y = \frac{3\sin 2x + 7}{3\sin x}$$ (4)

4.2 Determine the following limit:

$$\lim_{x \to \infty} \left(\frac{x^3 - 3x^2 + 9}{x^3 - 7} \right)$$ (3)

4.3 Simplify:

$$\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx$$ (3)

4.4 Determine:

$$\int_1^3 \left(\frac{5}{x} + 6 \right) dx$$ (3)

4.5 Integrate the following in terms of x:

$$\int \left(3\cos \pi x - 5\sqrt{x^3} + \frac{1}{\sqrt{x}} - 2^{-6x} - 5e^{-4x} + 7y \right) dx$$ (7)

[20]

QUESTION 5

5.1 The length of a rectangle is twice the breadth.

Calculate the length and the breadth of the rectangle if the area is:

$2^{-x+2} = 6^{2x-3}$ (5)

5.2 Solve for x:

$2^{-x+2} = 6^{2x-3}$ (3)

5.3 Given:

$$q = q_o\left(1 - e^{-\frac{t}{r}}\right)$$

Make t the subject of the formula. (3)

5.4 Given:

$$5a - 6b + 3c = -9$$
$$2a - 3b + 2c = -5$$
$$3a - 7b + 5c = -16$$

Use Cramer's rule to calculate for the value of a (5)

5.5 If:

$$\begin{vmatrix} 2 & -3 & -1 \\ -6 & 4 & 2 \\ 1 & 9 & 1 \end{vmatrix}$$

5.5.1 Write down and determine the minor of -3

5.5.2 Write down and determine the co-factor of 4

(2 × 2) (4)
[20]

TOTAL: 100

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MATHEMATICS N4

FORMULA SHEET

NEW SYLLABUS

$a^x = b \Leftrightarrow \log a^x = \log b$

$\ln x = \log_e x$

$\left(r\angle\underline{\theta}\right)^n = r^n\angle\underline{n\theta} \qquad a + bj = c + dj \Leftrightarrow a = c and b = d$

$\sin(a \pm b) = \sin a \cos b \pm \sin b \cos a$

$\cos(a \pm b) = \cos a \cos \mp \sin a \sin b$

$\sin^2 x + \cos^2 x = 1$

$1 + \cot^2 x = \cos ec^2 x$

$1 + \tan^2 x = \sec^2 x$

$$\tan(a \pm b) = \frac{\tan a \pm \tan b}{1 \mp \tan a \tan b}$$

y	$\frac{dy}{dx}$
an^n	nax^{n-1}
ka^x	$ka^x \ln a$
$k \ln x$	$\frac{k}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\cos ec^2 x$
$\sec x$	$\sec x \tan x$
$\cos ecx$	$-\cos ecx \cot x$

$$y = u(x) \cdot v(x)$$
$$\Rightarrow \frac{dy}{dx} = u(x)v^1(x) + u^1(x)v(x)$$

$$y = \frac{u(x)}{v(x)}$$
$$\Rightarrow \frac{dy}{dx} = \frac{v(x)u^1(x) - u(x)v^1(x)}{[v(x)]^2}$$

$$\frac{dy}{dx} = \frac{dy}{du} x \frac{du}{dx}$$

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + C$$

$$\int \frac{a}{x} dx = a\ell nx + c$$

$$\int ka^x dx = \frac{ka^x}{\ell na} + c$$

$$A_{ox} = \int_a^b y dx$$

$$\int \sin x dx = -\cos x + c$$

$$\int \cos x dx = \sin x + c$$

$$\int \tan x dx = \ell n \sec x + c$$

$$\int \sec x dx = \ell n(\sec x + \tan x) + c$$

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